

```

clc; clear; close all;
rb_bw = 180e3;
users = 1:5;
alloc_RB = [12 10 9 8 11];
mod_eff = [2 4 6 4 2];
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')

```

User Throughput (Mbps):

```
disp(throughput')
```

```

4.3200
7.2000
9.7200
5.7600
3.9600

```

```

figure
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```

